

Computational Physics, Homework 1:

Numerical differentiation, quadrature, and the BAO peak

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Summary

This assignment has three parts. The first two are numerical experiments on how truncation error and round-off error compete in single-precision (32-bit floating-point) arithmetic. I differentiated $\cos x$ and e^x at $x = 0.1$ and $x = 10$ with forward-, central- and extrapolated-difference formulas for step sizes h between 10^{-8} and 1, and I integrated $\int_0^1 e^{-t} dt$ with the midpoint, trapezoid and Simpson rules for $N = 2$ to 4×10^6 bins. In both experiments the relative error first follows the expected truncation power law (h , h^2 , h^4 for the derivatives; N^{-2} , N^{-4} for the integrals) and then turns over once round-off dominates. The best accuracy reached by each difference formula agrees with the standard estimate $\varepsilon_{\min} \sim \varepsilon_m^{n/(n+1)}$ for an n th-order formula: about 3–4, 5 and 6 significant digits for the first-, second- and fourth-order formulas, with $\varepsilon_m \simeq 1.2 \times 10^{-7}$. All three quadrature rules bottom out at the single-precision floor of $\sim 10^{-7}$ (seven digits); Simpson’s rule gets there with $N \approx 32$ bins whereas the second-order rules need a few hundred. Beyond $N \sim 10^5$ the error of a naive float32 running sum grows again, and faster than the $\varepsilon_m \sqrt{N}$ random-walk estimate, because the rounding errors of a smooth, monotonic integrand are correlated rather than random.

In the third part I transformed a tabulated Λ CDM matter power spectrum $P(k)$ into the two-point correlation function $\xi(r)$ by evaluating the spherical Fourier integral with a cubic spline in log–log space and composite Simpson quadrature on a k grid fine enough to resolve $\sin(kr)$. The baryon acoustic oscillation (BAO) bump in $r^2 \xi(r)$ peaks at $r = 105.7$ Mpc/ h , and its position is stable to better than 0.5 Mpc/ h against the choice of upper integration limit ($k_{\max} \geq 5$ h /Mpc), the k step, and the high- k taper used to suppress ringing.

Methods

Single-precision arithmetic and the error measure

Problems 1 and 2 are implemented in Python with NumPy. Every operand entering a difference formula or a quadrature sum is cast to `float32`, and the elementary functions are evaluated with `dtype=float32`, so every addition, subtraction, division and function evaluation rounds at the single-precision machine epsilon $\varepsilon_m = 2^{-23} \simeq 1.19 \times 10^{-7}$. The reference values (the analytic derivatives, and the exact integral $I = 1 - 1/e$) are computed in double precision, and the quantity plotted throughout is the relative error

$$\varepsilon = \frac{|\text{numerical} - \text{exact}|}{|\text{exact}|}.$$

Points where ε is exactly zero are omitted from the logarithmic plots.

Problem 1: numerical differentiation

Three finite-difference approximations to $f'(x)$ were coded:

$$\begin{aligned} D_F(h) &= \frac{f(x+h) - f(x)}{h}, & \text{truncation error } O(h), \\ D_C(h) &= \frac{f(x+h) - f(x-h)}{2h}, & O(h^2), \\ D_E(h) &= \frac{4D_C(h/2) - D_C(h)}{3}, & O(h^4), \end{aligned}$$

where the last is the Richardson extrapolation of the central difference, which cancels its leading h^2 error term. Each was applied to $\cos x$ and e^x at $x = 0.1$ and $x = 10$. The step size is sampled four times per octave, $h = 2^{-27}, 2^{-26.75}, \dots, 2^{0.75}$ (112 values from 7.5×10^{-9} to 1.7); h is rounded to `float32` before use and the same rounded value appears in the numerator and the denominator. Because the round-off error jumps erratically from one h to the next, the figure shows both the raw points (thin lines) and the median of ε in 28 logarithmic bins in h (thick lines). The binned median is the trend that should be compared with the estimates below; the single best point of a noisy curve is biased low by lucky cancellations.

The simple estimate for the attainable accuracy balances the truncation error $\sim h^n$ of an n th-order formula against the round-off error. Subtracting two nearly equal function values loses an absolute amount $\sim \varepsilon_m |f|$, and dividing by h gives a round-off contribution $\sim \varepsilon_m/h$ (up to the factor $|f/f'|$, which is ≈ 1 for e^x but $\cot 0.1 \approx 10$ for $\cos x$ at $x = 0.1$). The sum $h^n + \varepsilon_m/h$ is minimized at

$$h_{\text{opt}} \sim \varepsilon_m^{1/(n+1)}, \quad \varepsilon_{\text{min}} \sim \varepsilon_m^{n/(n+1)},$$

which for $n = 1, 2, 4$ predicts $h_{\text{opt}} \approx 3 \times 10^{-4}, 5 \times 10^{-3}, 4 \times 10^{-2}$ and $\varepsilon_{\text{min}} \approx 3 \times 10^{-4}, 2 \times 10^{-5}, 3 \times 10^{-6}$, i.e. about 3.5, 4.6 and 5.5 significant digits.

Problem 2: quadrature

With N equal bins of width $h = 1/N$ on $[0, 1]$ and nodes $t_i = ih$, the three composite rules are

$$\begin{aligned} I_{\text{mid}} &= h \sum_{i=0}^{N-1} f\left(t_i + \frac{h}{2}\right), & O(h^2), \\ I_{\text{trap}} &= \frac{h}{2} \left[f(t_0) + 2 \sum_{i=1}^{N-1} f(t_i) + f(t_N) \right], & O(h^2), \\ I_{\text{Simp}} &= \frac{h}{3} \left[f(t_0) + 4f(t_1) + 2f(t_2) + 4f(t_3) + \dots + 4f(t_{N-1}) + f(t_N) \right], & O(h^4), \end{aligned}$$

the last requiring even N . I used $N = 2^1, 2^2, \dots, 2^{22}$ (2 to 4 194 304). Because N is a power of two, h and every node ih are exactly representable in `float32`, so the node positions carry no rounding; all the round-off comes from evaluating e^{-t} and from the sum. The products of weights and function values are accumulated with a naive left-to-right loop in a `float32` accumulator (no pairwise or compensated summation), so the round-off behaviour is that of a plain hand-written loop. The plot includes reference lines $\propto N^{-2}$ and N^{-4} for the truncation regimes and $\varepsilon_m \sqrt{N}$, the growth expected if N independent rounding errors of size ε_m accumulated as a random walk.

Problem 3: the BAO peak in the correlation function

The correlation function is the spherical Fourier transform of the power spectrum,

$$\xi(r) = \frac{1}{2\pi^2} \int_0^\infty dk k^2 P(k) \frac{\sin(kr)}{kr}.$$

$P(k)$ is supplied as a table of 701 points, logarithmically spaced in k from 10^{-4} to 10^3 h/Mpc . Since $P(k)$ is close to a broken power law, I interpolated it with a cubic spline of $\ln P$ against $\ln k$ (`scipy.interpolate.CubicSpline`), which is far better behaved than a spline in linear variables. $P(k)$ is set to zero outside the tabulated range, consistent with $P(0) = 0$.

The integral was split at $k = 0.01$ h/Mpc . Below the split the integrand is smooth and a 601-point logarithmic grid suffices. Above it the factor $\sin(kr)$ oscillates with period $2\pi/r$, which at $r = 120$ Mpc/h is 0.052 h/Mpc , so I used a uniform grid with $\Delta k = 2 \times 10^{-4}$ h/Mpc up to $k_{\text{max}} = 10$ h/Mpc (about 5×10^4 points, at least 260 per oscillation). Composite Simpson's rule (`scipy.integrate.simpson`) was applied on each segment; $\sin(kr)/(kr)$ is evaluated as a sinc function so the $k \rightarrow 0$ limit is handled exactly.

The upper limit needs care. At large k the integrand $k^2 P(k)$ decays only roughly as $1/k$, so a hard cut at k_{max} introduces ringing in $\xi(r)$ with period $2\pi/k_{\text{max}}$. I tested robustness in two ways: by raising k_{max} from 1 to 20 h/Mpc , and by multiplying the integrand by a Gaussian taper $\exp[-(k/k_d)^2]$ with $k_d = 3$ h/Mpc . The taper removes the ringing while smoothing ξ only on scales $\sim 1/k_d \approx 0.3$ Mpc/h , far below the ~ 10 Mpc/h width of the BAO bump. The k step was also halved and more than doubled to check that the oscillations are resolved.

$\xi(r)$ was tabulated at 1401 points in $r \in [50, 120]$ Mpc/h ($\Delta r = 0.05$ Mpc/h). Over this range $r^2 \xi(r)$ is still decreasing steeply at $r = 50$ and has dropped again by $r = 120$, so the bump is an interior local maximum, not the largest value in the window. The peak is located as the most prominent interior maximum (`scipy.signal.find_peaks`) and its position refined by fitting a parabola through the three samples around it.

Results

Problem 1: differentiation

Figure 1 shows the four cases. Table 1 compares the simple estimates with what was measured; the measured columns give the range over the four (f, x) combinations.

Scaling and significant digits. On the large- h side of each minimum the curves run parallel to the h , h^2 and h^4 guides, confirming the order of each formula. The location of the minimum moves to larger h with increasing order, from a few times 10^{-4} (forward) to a few times 10^{-3} (central) to ~ 0.1 (extrapolated), in agreement with $h_{\text{opt}} \sim \varepsilon_m^{1/(n+1)}$. The extrapolated minimum sits a factor of 2–4 above the bare estimate because that estimate drops prefactors: the leading truncation term of D_E is $h^4 f^{(5)}/480$, and balancing this small coefficient against the round-off $\sim 3\varepsilon_m/h$ of the three combined differences gives $h_{\text{opt}} \approx 0.13$, inside the measured range. The error floors of the trend curves give roughly 3–4, 5 and 6 significant digits, bracketing the estimates of 3.5, 4.6 and 5.5. The spread within each method is the prefactor the estimate ignores: the forward difference of $\cos x$ at $x = 0.1$ is the worst case (ε floor 1.7×10^{-3}) because $|f/f'| = \cot 0.1 \approx 10$ amplifies the cancellation error tenfold, while e^x with $|f/f'| = 1$ reaches the estimate. The extrapolated formula does slightly better than its estimate, and individual raw points reach 7 digits for it, but this is the lucky-cancellation effect rather than a systematic gain.

Single-precision differentiation: relative error ε vs step size h

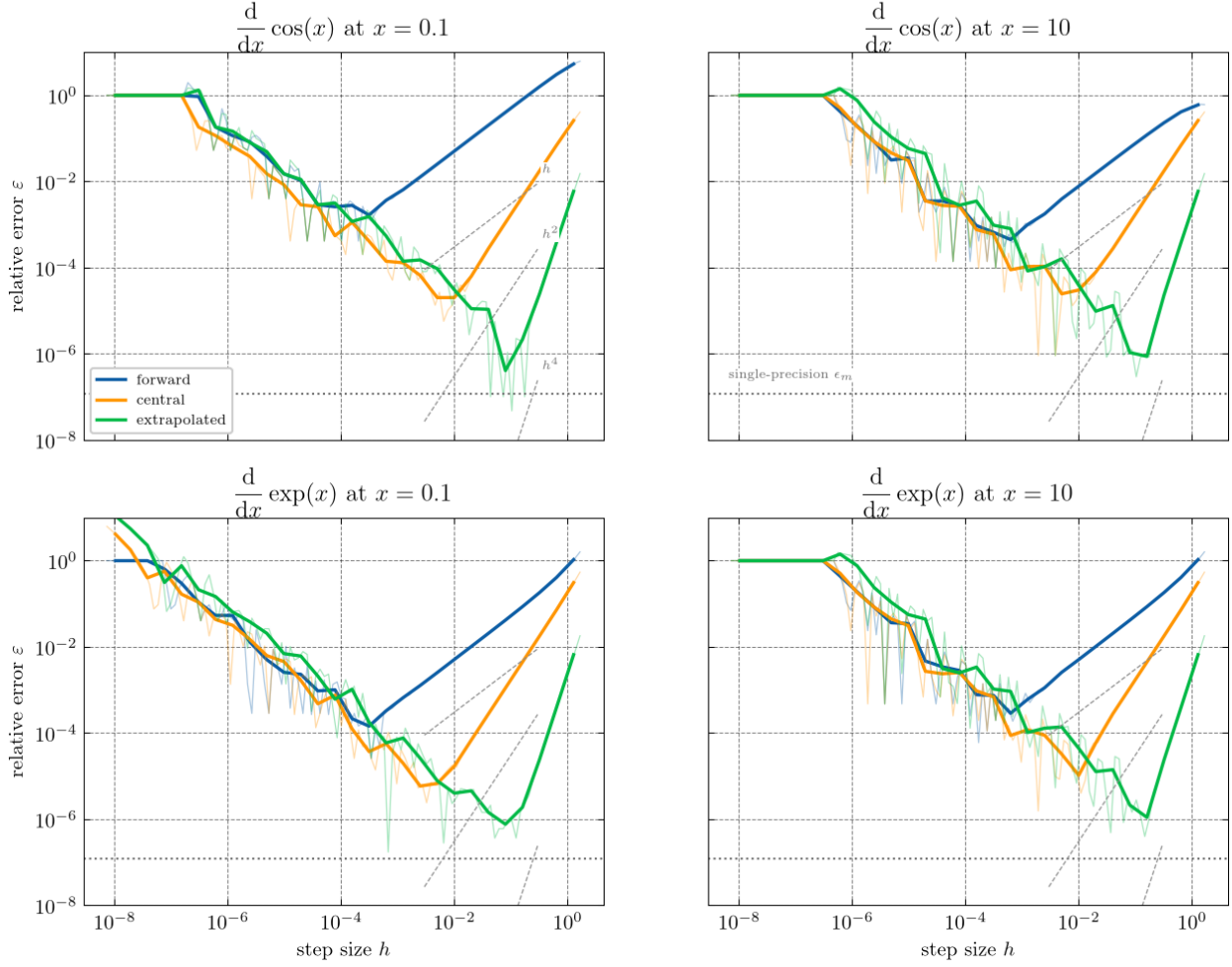


Figure 1: Relative error ε of the single-precision forward (blue), central (orange) and extrapolated (green) difference approximations to the derivative of $\cos x$ (top) and e^x (bottom) at $x = 0.1$ (left) and $x = 10$ (right), as a function of the step size h . Thin lines are the raw results at 112 step sizes; thick lines are the median in logarithmic bins in h . The dashed grey guides have slopes h , h^2 and h^4 , and the dotted horizontal line marks the single-precision machine epsilon $\varepsilon_m = 1.2 \times 10^{-7}$. For every method the error decreases with the expected power of h on the right (truncation regime), passes through a minimum, and rises as $1/h$ with large scatter on the left (round-off regime) until it saturates at $\varepsilon = 1$ where $f(x+h)$ and $f(x)$ become the same `float32` number.

The truncation and round-off regimes. Three regimes are visible in every panel of Fig. 1:

- **Large h (right of the minimum): truncation.** The error is the neglected Taylor term, so it is smooth, deterministic and falls as h , h^2 or h^4 . The crossover to this regime moves to the right for higher order, which is why the extrapolated formula is both the most accurate and the least demanding on the choice of h .
- **Small h (left of the minimum): round-off.** $f(x+h)$ and $f(x)$ agree to more and more digits, the subtraction keeps only the disagreeing ones, and the result is divided by h , so the error rises as ε_m/h , one decade per decade. It is noisy because the cancellation error depends on the last bits of the rounded function values, which change erratically with h . All three

Table 1: Predicted and measured optimum step size and best relative error for the three difference formulas in single precision. “Trend” rows refer to the binned-median curves of Fig. 1; “raw best” rows give the single lowest raw point, which is systematically lower because it selects lucky cancellations. Measured rows give the range over the four (f, x) cases.

method		h at minimum	$\varepsilon_{\min}$	digits
forward, $O(h)$	estimate	3.4×10^{-4}	3.4×10^{-4}	3.5
	trend	3×10^{-4} – 6×10^{-4}	1.4×10^{-4} – 1.7×10^{-3}	2.8–3.8
	raw best	2.6×10^{-5} – 3.5×10^{-4}	2.6×10^{-5} – 4.2×10^{-4}	3.4–4.6
central, $O(h^2)$	estimate	4.9×10^{-3}	2.4×10^{-5}	4.6
	trend	2.5×10^{-3} – 1.0×10^{-2}	5.9×10^{-6} – 2.5×10^{-5}	4.6–5.2
	raw best	2.0×10^{-3} – 6.6×10^{-3}	9.1×10^{-7} – 5.7×10^{-6}	5.2–6.0
extrapolated, $O(h^4)$	estimate	4.1×10^{-2}	2.9×10^{-6}	5.5
	trend	0.08–0.16	4.1×10^{-7} – 1.1×10^{-6}	6.0–6.4
	raw best	6.9×10^{-4} –0.15	4.8×10^{-8} – 4.0×10^{-7}	6.4–7.3

formulas suffer equally here, since they differ only in how the same cancelling differences are combined.

- **Very small h : total loss.** For $h \lesssim 2 \times 10^{-7}$ at $x = 0.1$ and $h \lesssim 5 \times 10^{-7}$ at $x = 10$ (where $x + h$ itself rounds back to x), $f(x \pm h)$ is the same `float32` number as $f(x)$, the numerator is exactly zero and the relative error saturates at $\varepsilon = 1$: the flat segments at the far left.

Problem 2: quadrature

Figure 2 shows three regimes.

- **Small N : truncation.** The midpoint and trapezoid errors fall as N^{-2} and lie almost on top of each other, as expected since their leading error terms differ only by a factor of $-1/2$. Simpson’s rule falls as N^{-4} and is already at 10^{-7} by $N = 32$. This regime is smooth and deterministic.
- **Intermediate N : the single-precision floor.** Once the truncation error drops below the precision of the accumulated sum, every rule flattens out at $\varepsilon \sim \varepsilon_m \approx 10^{-7}$, about seven significant digits, and scatters between 10^{-8} and 10^{-6} from one N to the next. Simpson’s rule hits the floor at $N \approx 30$, the second-order rules at $N \approx 300$ –500. Adding bins beyond this point buys nothing.
- **Large N : round-off growth.** Past $N \sim 10^5$ the error rises again, reaching 3×10^{-3} to 2×10^{-2} at $N = 4 \times 10^6$, ten to a hundred times above the $\varepsilon_m \sqrt{N}$ line. The nodes are exact for power-of-two N , so this growth comes entirely from the running sum: the accumulator grows to order N while each term is of order one, so the rounding error of each addition grows in proportion to the partial sum, and because the integrand varies slowly, consecutive additions round almost identically: the per-addition errors keep the same sign over runs of hundreds to 10^6 terms and add coherently rather than cancelling as a random walk. I confirmed this by summing the identical `float32` function values in double precision (or with pairwise summation in `float32`), which keeps the error at 10^{-8} all the way to $N = 4 \times 10^6$.

The best accuracy is essentially the same for all three rules (Table 2), about seven to eight digits; the individual points below ε_m are fortunate roundings rather than a real gain. What distinguishes

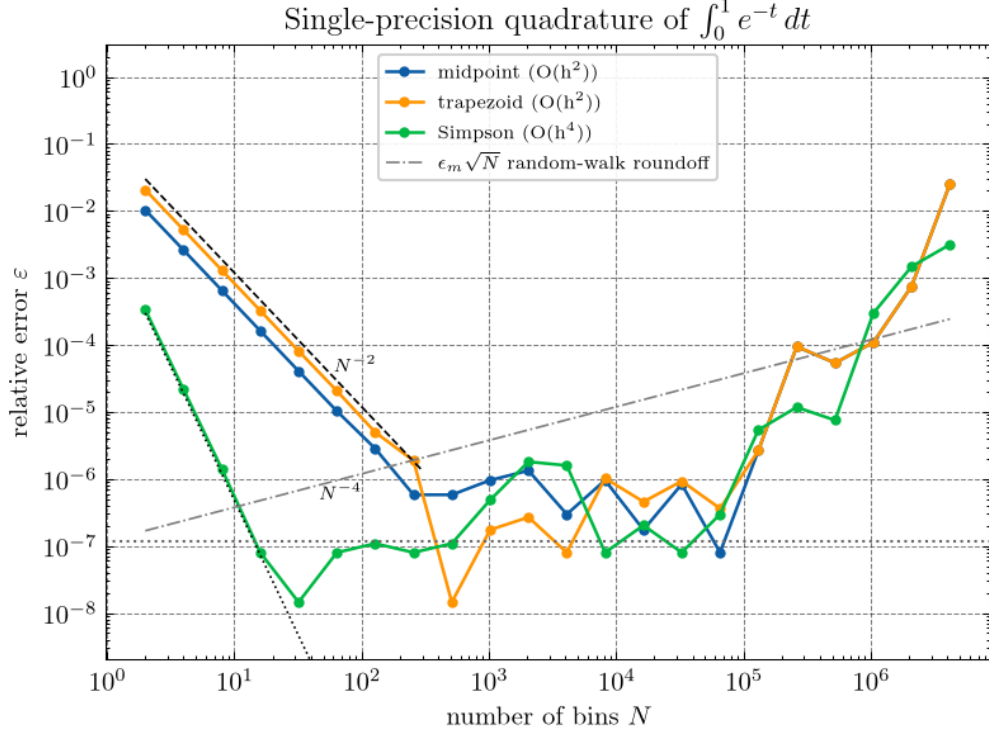


Figure 2: Relative error of the single-precision midpoint (blue), trapezoid (orange) and Simpson (green) rules for $I = \int_0^1 e^{-t} dt$ as a function of the number of bins $N = 2^1, \dots, 2^{22}$. The dashed and dotted black lines are $\propto N^{-2}$ and N^{-4} ; the dash-dotted grey line is $\epsilon_m \sqrt{N}$, the random-walk estimate for accumulating N rounding errors; the dotted horizontal line is ϵ_m . The two second-order rules coincide in the truncation regime and reach the $\sim 10^{-7}$ floor near $N \approx 300$, Simpson’s rule reaches it by $N \approx 30$, and all three rise again past $N \sim 10^5$, more steeply than $\sqrt{N}$, because the `float32` running sum loses precision as it grows.

the methods is the cost of getting there: Simpson’s rule needs an order of magnitude fewer function evaluations than the second-order rules.

Table 2: Number of bins and relative error at the minimum of each curve in Fig. 2. The exact value is $I = 1 - 1/e = 0.6321205588$.

method	order	N at minimum	minimum ϵ
midpoint	$O(h^2)$	65 536	8.0×10^{-8}
trapezoid	$O(h^2)$	512	1.5×10^{-8}
Simpson	$O(h^4)$	32	1.5×10^{-8}

Problem 3: the BAO peak

Figure 3 shows the input power spectrum, the baryon wiggles isolated by dividing out a smooth fit, and the resulting $r^2 \xi(r)$. The correlation function decreases from $r^2 \xi \approx 26$ (Mpc/h)² at $r = 50$ Mpc/h to a trough of ≈ 7.5 at $r \approx 86$ Mpc/h, rises to the BAO bump, and falls again toward zero by $r = 120$ Mpc/h.

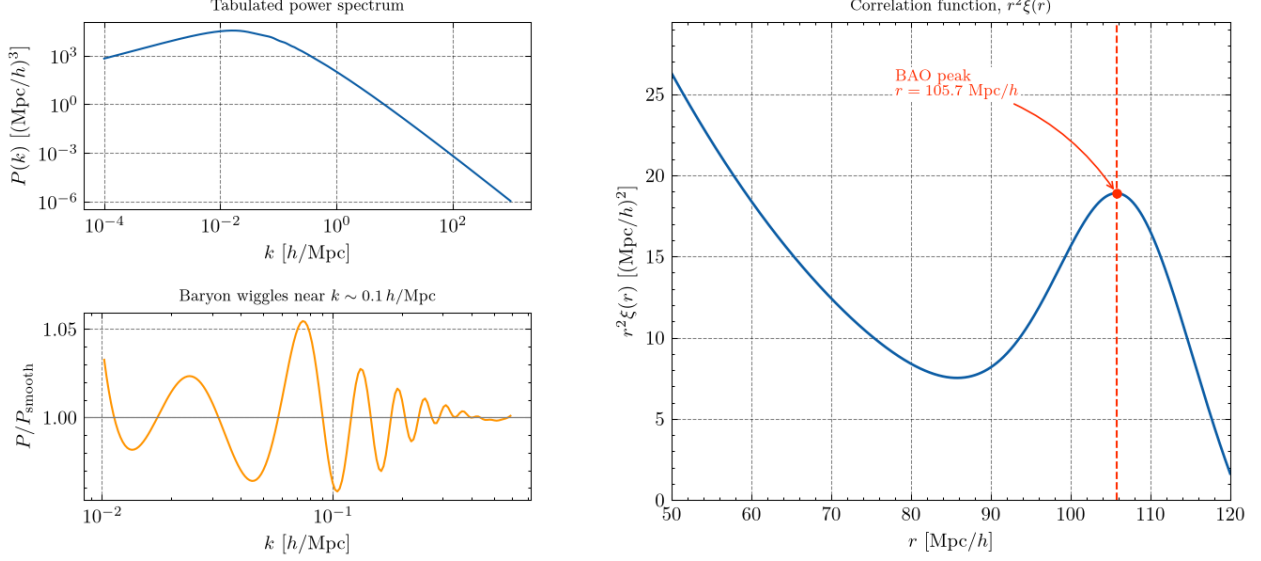


Figure 3: Left, top: the tabulated linear matter power spectrum $P(k)$ at $z = 0$. Left, bottom: $P(k)$ divided by a smooth sixth-order polynomial fit in $\ln k - \ln P$ over $0.01 < k < 0.6 \text{ h/Mpc}$, which exposes the baryon wiggles, a damped oscillation of a few per cent amplitude around $k \sim 0.1 \text{ h/Mpc}$. Right: the correlation function multiplied by r^2 over $50 \leq r \leq 120 \text{ Mpc/h}$, computed with $k_{\text{max}} = 10 \text{ h/Mpc}$, $\Delta k = 2 \times 10^{-4} \text{ h/Mpc}$ and a Gaussian taper at $k_d = 3 \text{ h/Mpc}$. The baryon wiggles in k space appear as a single bump in r space; its maximum, the BAO peak, is marked by the red point and dashed line at $r = 105.7 \text{ Mpc/h}$.

Table 3: Position of the BAO peak for different upper limits k_{max} , k steps and taper scales k_d (“none” is a hard cut at k_{max}).

$k_{\text{max}} \text{ [h/Mpc]}$	$\Delta k \text{ [h/Mpc]}$	$k_d \text{ [h/Mpc]}$	$r_{\text{peak}} \text{ [Mpc/h]}$
1	2×10^{-4}	none	103.77
2	2×10^{-4}	none	105.26
5	2×10^{-4}	none	106.18
10	2×10^{-4}	none	105.87
20	2×10^{-4}	none	105.71
10	1×10^{-4}	3	105.73
10	5×10^{-4}	3	105.73
10	2×10^{-4}	1	105.69
10	2×10^{-4}	3	105.73
20	2×10^{-4}	3	105.73

Robustness of the upper limit. Table 3 lists the peak position for the tested settings. With a hard cut, truncating at $k_{\text{max}} = 1 \text{ h/Mpc}$ is visibly not converged (the peak moves by 2 Mpc/h), but for $k_{\text{max}} \geq 5 \text{ h/Mpc}$ the position is stable to better than 0.5 Mpc/h , with the residual wander caused by ringing of period $2\pi/k_{\text{max}}$. Once the Gaussian taper is applied, the result is identical to 0.01 Mpc/h whether $k_{\text{max}} = 10$ or 20 h/Mpc , whether Δk is 10^{-4} , 2×10^{-4} or $5 \times 10^{-4} \text{ h/Mpc}$, and changes by only 0.04 Mpc/h if the taper is moved to $k_d = 1 \text{ h/Mpc}$. The adopted setting, $k_{\text{max}} = 10 \text{ h/Mpc}$ with $\Delta k = 2 \times 10^{-4}$ and $k_d = 3 \text{ h/Mpc}$, is therefore converged.

Result. The BAO peak is at

$$r_{\text{peak}} = 105.7 \text{ Mpc}/h, \quad r^2 \xi(r_{\text{peak}}) = 18.9 (\text{Mpc}/h)^2,$$

with a numerical uncertainty well below $0.5 \text{ Mpc}/h$. This is the scale of the sound horizon at the baryon-drag epoch imprinted in the matter distribution, and it lies in the $\sim 100\text{--}110 \text{ Mpc}/h$ range expected for a standard ΛCDM cosmology. The tabulated $\xi(r)$ is saved alongside the code.

Code

All code, the input power spectrum, the figures and the numerical outputs are in the public GitHub repository

<https://github.com/hz-xiaxz/compphys-hw1>

The scripts are plain Python (NumPy, SciPy, Matplotlib, SciencePlots) managed with `uv`:

- `problem1.py`: forward, central and extrapolated differences in single precision; writes `figs/problem1_error.png` and prints Table 1’s raw minima.
- `problem2.py`: midpoint, trapezoid and Simpson rules with a naive `float32` accumulator; writes `figs/problem2_error.png`.
- `problem3.py`: log–log spline of $P(k)$, the split Simpson integration, the convergence scan of Table 3, the peak finder, and `figs/problem3_bao.png`; also writes $\xi(r)$ to `results/xi_of_r.txt`.
- `plotstyle.py`: shared figure style. `results/` holds the printed output of each script, and `writeup/` holds the source of this document.

Running `uv sync` followed by `uv run python problemN.py` reproduces every figure and number quoted here.